



Course Number: VGP240
Course Title: 3D Graphics and Applications

Assignment: Vector and Pix Editor

Date Assigned: Week 1

Due Date: Week 2

Description:

There are two parts to this assignment. Part A will require you to start implementing your own math library in Pix by adding structs for Vector2 and Vector3. Part B will familiarize you with using Pix's scripting system.

Part A:

Add struct Vector2 and Vector3:

- Add a new project filter in Pix and all it **Math**
- Add new header files Vector2.h and Vector3.h
- Add structs Vector2 and Vector3 to the corresponding files
- The structs should have float members x, y, and x, y, z
- Add operator overloads for both structs (e.g. +, -, *, /)
- Add new header file MathHelper.h
- In MathHelper.h, you should include both vector header files
- In MathHelper.h add the following helper functions:

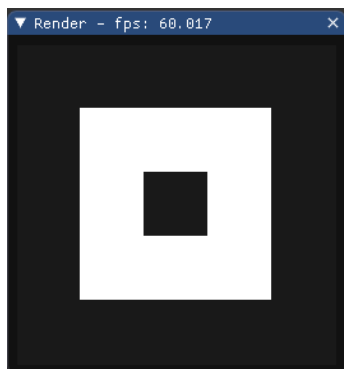
```
float MagnitudeSquared(Vector2 v);  
float MagnitudeSquared(Vector3 v);  
float Magnitude(Vector2 v);  
float Magnitude(Vector3 v);  
Vector2 Normalize(Vector2 v);  
Vector3 Normalize(Vector3 v);  
float Dot(Vector2 a, Vector2 b);  
float Dot(Vector3 a, Vector3 b);  
Vector3 Cross(Vector3 a, Vector3 b);
```

- Add new source file MathHelper.cpp and add the function implementations

Part B:

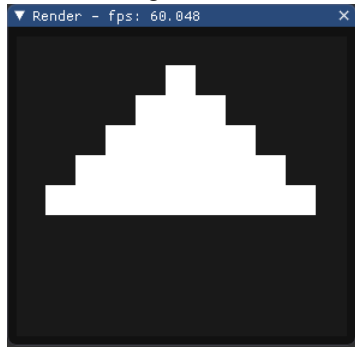
Using the Pix commands **SetResolution** and **DrawPixel**, create the following scripts:

- 1) Draw a box with resolution 5x5 with pixel size 50





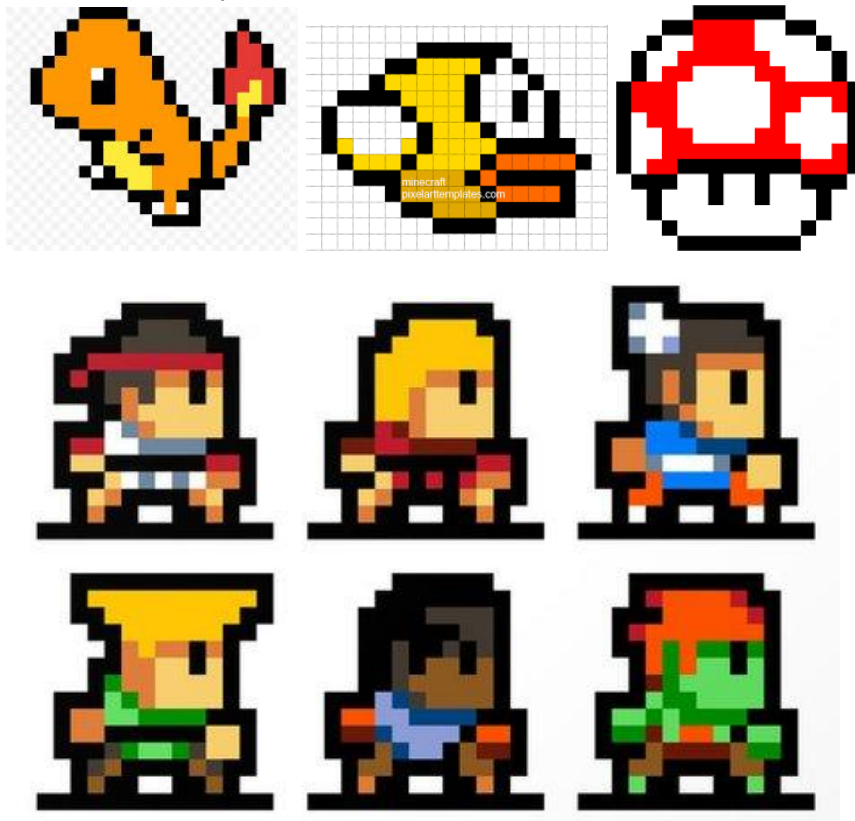
- 2) Draw a triangle with resolution 11x10 with pixel size 25



(triangle.pix)

Using CmdDrawPixel as a reference, add a new Pix command **SetColor**. Then add a new script of your own creation to draw a colored image of something iconic in games. Your image needs to have a minimum of resolution 10x10 with pixel size 25.

Here are some inspirations:



Deliverables:

You should have all your code implementations ready for evaluation and demo in class. Instructions will follow with hand in procedure.